



Plotting and interpreting linear functions

Teacher resource

Question 1

The cost in dollars, K , to print n books is given by the rule $K = 500 + 10n$.

A line graph is drawn to show the relationship between the number of books printed and the cost of printing the books.

Which point on the graph would indicate the cost of printing 100 books?

- ☐ (100, 500)
- ☐ (100, 1010)
- ☐ (100, 1500)
- ☐ (100, 5000)

Answer: C

Skill: Students understand that a linear graph is a set of points that satisfy a linear function.

Background information

Students should be encouraged to see the links between a linear graph, the relationship between the variables, and the features of a table of values for a linear function.

When constructing a linear graph 'by hand', students may need prompting to first construct a table of values and then use the maximum values in this table to inform their choice of scale on the axes. They may also need to be encouraged to use multiplication and division to choose the most appropriate scale rather than trial and error with addition.

Additional questions

1. What point on the graph would indicate the cost of printing 500 books?
2. What is the maximum number of books that could be printed for \$1000?
3. Each symbol and number in the rule $K = 500 + 10n$ has its own significance. In the rule, what does the 10 mean? What could be the meaning of the 500?

Question 2

The fare for a trip with *Fabulous Taxis* is given as \$4.00 flagfall plus \$2.00 per kilometre.

A line graph showing the relationship between the distance travelled, in kilometres, and the total cost of the trip, in dollars, is drawn.

Which statement about the graph would be true?

- ☐ The graph meets the vertical axis at the point (0, 2).
- ☐ The graph meets the vertical axis at the point (0, 4).
- ☐ The point (2, 4) lies on the graph.
- ☐ The point (4, 2) lies on the graph.

Answer: B

Skill: Students understand that a linear graph is a set of points that satisfy a linear function.

Additional questions

1. What is the fare for a trip of 25 kilometres with *Fabulous Taxis*?
2. What is the maximum distance that could be travelled for \$50?
3. If b is the number of kilometres travelled and m is the number of dollars in the fare, what is the relationship between b and m ?

Question 3

A graph of the equation $y = 2x - 5$ forms a straight line.

The line can be drawn by joining two points.

Choose two appropriate points to draw this line.

- ☐ (5, 2) and (2, 5)
- ☐ (2, 5) and (-1, -7)
- ☐ (3, 1) and (5, 2)
- ☐ (-1, -7) and (3, 1)

Answer: D

Skill: Students check whether points belong to a line graph.

Additional questions

1. What is the relationship between the 2 in this equation and the graph of the function?
2. What is the relationship between the -5 in this equation and the graph of this function?
3. Where will this line cut the horizontal axis?

Question 4

Choose the graph that has the greatest gradient.

- ☐ $y = 5x + 9$
- ☐ $5y = 9x + 2$
- ☐ $5y = 10x + 9$
- ☐ $y = 8x - 2$

Answer: D

Skill: Students interpret a linear equation.

Additional questions

1. What is the gradient of the graph of equation B?
2. Are any of the four graphs parallel?

Student activity: Plotting and interpreting linear functions

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- ☐ The point (2, 4) lies on the graph.
- ☐ The point (4, 2) lies on the graph.

Question 3

A graph of the equation $y = 2x - 5$ forms a straight line. The line can be drawn by joining two points.

Choose two appropriate points to draw this line.

- ☐ (5, 2) and (2, 5)
- ☐ (2, 5) and (-1, -7)
- ☐ (3, 1) and (5, 2)
- ☐ (-1, -7) and (3, 1)

Question 4

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- ☐ $5y = 9x + 2$
- ☐ $5y = 10x + 9$
- ☐ $y = 8x - 2$